# Nov 3 — 3 papers

**Episode 46** · November 3, 2026 · 3 papers

## Understanding LLM-Based Bargaining of Driving Agents in Mixed-Autonomy Traffic

SKIM

Lingxiao Zhou, Di Chen, Jia Li

*They have LLM agents playing autonomous and human drivers haggle in text over shared road space, then label every bid and tone shift to see which interaction patterns lead to deals and which lead to breakdowns.*

**PROBLEM** LLM agents are increasingly used to simulate social behaviour, but nobody has looked closely at how an LLM-driven negotiation over road space between automated and human drivers actually plays out turn by turn, as opposed to just whether it ends in agreement.

**METHOD** They run controlled bargaining simulations between AV and HV agents across populations of agents. They vary three factors: traffic density, each agent's objective, and whether the agent is given traffic-domain knowledge. They then label each turn with a taxonomy covering two things, the numerical bid and the communicative attitude, and they compare each agent's private reasoning with the message it actually sends.

**RESULTS** The abstract gives no numbers. It reports only qualitative patterns: agreement rates differ across conditions, deals show more constructive back-and-forth, breakdowns show more escalation, and the tone an agent sends often differs from its private stance even when its bid stays the same.

**CAVEAT** Nothing ties the bargaining to vehicle dynamics, safety, or timing, so an agreement is an outcome of text generation and not a maneuver that was ever executed or checked for feasibility. It also models human drivers with an LLM without validating that against real driver behaviour. The links between behaviour patterns and outcomes are descriptive correlations from one LLM setup, and as far as the abstract shows there is no test across models or prompts.

**WHY IT MATTERS** This is peripheral to your work. It shares the word negotiation, but it has no message protocol, no latency or packet loss, no trajectories, and no real conflict resolution. It is closer to computational social science than to maneuver coordination. The one useful idea is the gap between private intent and public message, which matters if you ever think about strategic or untruthful intent sharing. Otherwise, cite it as an example of LLM-based negotiation and move on.

arXiv:2610.41239v1 PDF incremental

## Interaction-Aware Sampling-Based Motion Planning for Autonomous Surface Vessels in Urban Waterways

SKIM

Elia Trevisan et al.

*A Delft-style MPPI planner for small autonomous boats in Amsterdam's canals that plans its own path while also predicting the other boats' paths, with no communication between vessels.*

**PROBLEM** Vessels in narrow canals must share space and follow rules such as keeping to starboard, but they usually don't know each other's intentions. Planners that predict the other vessels first and then plan the ego vessel ignore the fact that the other vessels react to the ego. Rule costs are also discontinuous, which gradient-based optimizers handle poorly.

**METHOD** Each vessel runs model predictive path integral control, which is sampling-based MPC, over a joint state. Each sample rolls out the ego trajectory together with predicted trajectories for nearby vessels, so the prediction and the plan are shaped by one shared cost. Traffic rules enter that cost as discontinuous penalties, which sampling handles without smoothing.

**RESULTS** The abstract reports 780 randomized simulation runs on real Amsterdam canal maps. These cover two-vessel encounters and four-vessel scenarios, and every run reached its goal with no collisions or deadlocks. The method beats an optimization-based planner built for the same domain and an ablation that drops the interaction. On hardware, two scaled prototypes ran planning and localization onboard at 10 hertz and resolved head-on and crossing encounters without collisions or rule violations. The abstract gives no margins for the comparison with the baselines.

**CAVEAT** The zero-deadlock claim probably depends on every simulated vessel running the same planner with the same cost model. In that setting the joint prediction effectively assumes a shared, rule-following policy, which works in your favour. A heterogeneous or adversarial traffic mix is the real test, and the abstract doesn't say whether it was run. The hardware validation covers only two vessels in two encounter types. Scenarios with four agents is a small scale for a claim about urban traffic.

**WHY IT MATTERS** This sits next to your work rather than in it. It is inference-based interaction-aware planning with no message exchange, which is exactly the alternative your negotiation research is meant to improve on. It is useful as a strong no-communication baseline, and as evidence that encoding right-of-way rules as discontinuous costs in sampling-based MPC removes deadlock in symmetric encounters. It says nothing about protocols, message loss, partial participation, or platooning.

arXiv:2610.41676v1 PDF incremental

## Learning Side-by-Side Navigation for Personal Mobility Vehicles from Human Drivers

SKIM

Luis Yoichi Morales, Florent Ferreri

*Two indoor personal mobility vehicles, which are small single-seat vehicles, learn from human drivers to travel side by side so their riders can talk, using a shared decision rule over each other's candidate trajectories, and are tested only in a kinematic simulator.*

**PROBLEM** Keeping two autonomous vehicles in a social side-by-side formation along narrow, cluttered corridors needs joint decisions about lanes and speed, and hand-tuned formation controllers neither reproduce how people actually ride together nor adapt the usable lanes to the geometry. The authors treat this as a two-agent coordination problem with no explicit negotiation channel.

**METHOD** Each vehicle runs its own sampling-based model-predictive planner, and the only thing exchanged is odometry each vehicle could observe of the other. Each vehicle scores every pair of candidate trajectories, which gives two utility matrices, and a common resolution rule picks the joint choice, so this is effectively a fixed convention over a bimatrix game rather than a proposal and acceptance exchange. Cost weights are fitted by inverse reinforcement learning from 409 minutes of ten human pairs. Usable lanes are derived offline from corridor geometry. A hard safety layer only allows a move if the vehicle can still brake clear, measured from its whole body, and a stuck vehicle reverses along its own path.

**RESULTS** Across 24 simulated runs in the recorded conditions and 24 in four generated obstacle layouts, no vehicle hit a wall or cone. Among obstacles the pair held conversational formation 60.5 and 68.9 percent of the time, anticlockwise and clockwise, against 34.0 and 36.0 percent for the human drivers. In a clear aisle it reached 81.2 and 91.5 percent against 52.9 and 56.0 percent, at matched speed and with much lower variance.

**CAVEAT** Everything is in a differential-drive kinematic simulator, with no hardware run, no sensing noise, and no test where the shared odometry is delayed or lost. Collision results cover only walls and cones, so vehicle-to-vehicle near misses and pedestrians go unreported. The headline comparison also undercuts itself. A policy learned by inverse reinforcement learning that beats its own demonstrators on formation time, and settles on one lateral formation while the humans switched between diagonal and abreast, has not reproduced human behaviour. It may simply be optimising the formation metric harder. There is no ablation against hand-tuned weights or against a version without the shared resolution rule, so we cannot tell what the learning contributes.

**WHY IT MATTERS** This is mostly peripheral to your work. It is two indoor vehicles at walking speed, with no message design, no proposal or acceptance scheme, no partial participation, and no road maneuvers. The one useful idea is the coordination mechanism itself. Both agents evaluate the same joint trajectory-pair space and apply a common deterministic rule, which gives a minimal baseline for implicit agreement without intent exchange. You could cite it as the no-communication end of the spectrum that your intent-sharing protocols should beat, especially for deadlock and yielding cases.

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